

SANJAI CRV

+91 63839 21051 | sanjaicrv05@gmail.com | linkedin.com/in/sanjai-crv | github.com/sanjaicrv | LeetCode

Summary

Software Engineer with strong expertise in **Java, Spring Boot, REST APIs, and full-stack development**. Proven ability to design scalable backend systems, optimize database performance, and apply data structures & algorithms to real-world problems. Solid understanding of **OS, DBMS, CN, and System Design** fundamentals. Actively solving 300+ LeetCode problems with a FAANG-pattern focus; passionate about writing clean, high-impact code at scale.

Education

Sri Krishna College of Technology, Coimbatore

CGPA: 8.1 / 10

B.Tech in Information Technology

Technical Skills

- **Languages & Frameworks:** Java, Python, C++, SQL | Spring Boot, React.js, REST APIs, MVC Architecture
- **Tools & Cloud:** Git, GitHub, Docker, Postman, Swagger UI | MySQL, AWS (S3, EC2)
- **Core CS:** Data Structures & Algorithms, OOP, Low-Level Design, System Design, OS, DBMS, CN
- **AI / ML:** YOLOv8, OpenCV, Image Processing, Data Augmentation

Projects

Personal Budget Planner *Spring Boot · React.js · MySQL · REST APIs · Swagger UI* **2025**

- Engineered a production-grade full-stack budget platform with **12+ RESTful API endpoints** covering income tracking, expense categorization, and monthly analytics — handling **500+ financial transactions** in test load.
- Reduced MySQL query execution time by **40%** through strategic indexing, query restructuring, and JOIN optimization on normalized relational schemas aligned with **DBMS normalization principles**.
- Achieved **100% API documentation coverage** via Swagger UI, eliminating manual spec overhead and enabling faster frontend-backend integration.

Frontend Repo | Backend Repo | Live Demo

Parking Lot System — Low-Level Design *Java · OOP · Design Patterns · DSA* **2026**

- Architected a scalable system adhering to **SOLID principles** with **Strategy + Factory design patterns**, supporting 3 vehicle categories and multi-floor layouts without modifying existing code.
- Achieved **O(1) slot allocation/deallocation** via a HashMap-based index — **10× faster** than O(n) scan, validated at **1,000+ concurrent simulated operations**.
- Modeled **6 core domain entities** across 3 clean architectural layers, enabling fully unit-testable components and straightforward feature extension with zero regression risk.

GitHub Repo | Live Demo

Jewellery Defect Detection System *YOLOv8 · OpenCV · Python · Computer Vision* **Ongoing**

- Developing an AI-based jewellery defect detection system using **YOLOv8** and **OpenCV** for automated visual inspection in manufacturing workflows.
- Completed image preprocessing and dataset preparation using augmentation techniques such as mosaic augmentation and HSV transformation to improve model robustness.
- Currently working on YOLOv8 model training, defect classification, and real-time inference pipeline development for defect localization and detection.

GitHub Repo

Achievements & Activities

- **Participant** — Smart India Hackathon 2024: Developed an AI-powered **Gunshot Detection System** using sound localization and machine learning concepts for real-time threat detection.
- **4th Place** — Netrix'25 Codethon at KPR Institute: Ranked among the top teams out of **100+ participants** in a competitive algorithmic coding contest.
- **Finalist** — Yugam 2025 Code Event: Qualified through multiple competitive programming rounds focused on **Data Structures & Algorithms**.
- **Technical Presenter** — Presented an AI-based **Travel & Tourism** project at KPR Symposium, demonstrating practical applications of recommendation systems and web technologies.
- **Participant** — NLP Workshop at **Microsoft, Bangalore** (2023): Gained exposure to Natural Language Processing workflows, transformer models, and AI applications.
- **300+ LeetCode Problems Solved** — Consistent practice across core **Data Structures & Algorithms** topics including Arrays, Trees, Graphs, Dynamic Programming, Backtracking, and Sliding Window techniques with preparation focused on FAANG-level coding interviews.